Clothing Segmentation with YOLOv8

1. Introduction

Objective: The goal of this task is to perform image segmentation of clothing items (full sleeve, half sleeve, pants) using the YOLOv8 segmentation model.

Why YOLOv8?

(1) Fast and efficient real-time object detection and segmentation.
(2) Handles multiple classes simultaneously.
(3) Easy to fine-tune on custom datasets.

2. Dataset Preparation

Dataset was annotated with three classes: [I used Make sense in order to Annotate image Webiste Link]

- full_sleeve
- half_sleeve
- pants_full

3. Dataset organized into the YOLO format:

```
dataset/
├── images/
│   ├── train/
│   └── val/
├── labels/
│   ├── train/
│   └── val/
└── data.yaml
```

data.yaml file specified dataset location and class names.

4. Implementation Steps

Step 1: Install Dependencies

!pip install ultralytics

Step 2: Dataset Verification

!ls dataset/images/train | head -5
!ls dataset/images/val | head -5

Step 3: Train YOLOv8 Segmentation Model

4. Results

- Training Progress
- The model successfully trained for 30 epochs.
- Training logs saved under runs/seg/cloth_seg_run.

Visualization

Example segmented outputs (clothing areas highlighted with masks) are available in the notebook.

6. Conclusion

- Successfully trained a YOLOv8 segmentation model on a custom clothing dataset.
- The model is able to detect and segment clothing types (full_sleeve, half_sleeve, pants_full).